

B.Tech Examination-2023
Electronics and Communication Engineering
(Odd Semester Regular and Supplementary)
(For MEN and CEN)
Basic Electronics Engineering (ECEUGES01)

Full Marks: 80

Time: 3.00 Hrs

- Answer question no.1 and seven questions from rest of the questions.
- Answer all parts of a question in same place.
- Figures on the right-hand side margin indicate full marks.

1. Choosing the correct alternative from the given options answer any ten questions.

10

- (a) The sharing of electrons between neighboring atoms form
 - i) ionic bonds ii) covalent bond iii) metallic bonds iv) van der Waals bonds
- (b) When trivalent impurities are added to an intrinsic semiconductor, it becomes
 - i) P-type semiconductor ii) N-type semiconductor iii) PN-junction iv) all of these
- (c) The width of the depletion layer of a PN-junction is
 - i) independent of applied voltage ii) increased with applied forward bias iii) increased with applied reverse biased iv) increased with doping
- (d) Zener diode is used as
 - i) a voltage regulator ii) a rectifier iii) an amplifier iv) a multiplier
- (e) In a transistor, the reverse saturation current I_{CO} becomes double for each
 - i) 1^0 rise of temperature ii) 5^0 rise of temperature iii) 10^0 rise of temperature iv) 20^0 rise of temperature
- (f) When a transistor operates in a cut-off region, it behaves as
 - i) a closed switch ii) an open switch iii) an amplifier iv) all of these
- (g) A FET is
 - i) a voltage-controlled device ii) a current-controlled device iii) a resistance-controlled device iv) an impedance-controlled device
- (h) When a JFET operates above pinch-off voltage,
 - i) drain current remains constant ii) drain current increases rapidly iii) drain current decreases gradually iv) depletion region becomes zero
- (i) The maximum rate of change of output voltage per unit time is
 - i) slew rate ii) CMRR iii) offset voltage iv) supply-voltage rejection ratio
- (j) The feedback element in an integrator is
 - i) a diode ii) a capacitor iii) a resistor iv) an inductor
- (k) Negative feedback in amplifier
 - i) increases bandwidth and increases gain ii) increases bandwidth and decreases gain iii) decreases bandwidth and decreases gain iv) decreases bandwidth and increases gain
- (l) Which topology of feedback amplifier has high input and output impedance?
 - i) Voltage-series ii) Voltage-shunt iii) Current-series iv) Current-shunt

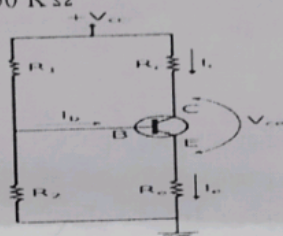
2. (a) Define extrinsic semiconductors. How can they be formed?
- (b) Prove that conductivity of a semiconductor is $\sigma = ne\mu_c + pe\mu_v$
- (c) Write the Fermi-Dirac Distribution Function and define Fermi level.

4
4
2

3. (a) Draw the circuit diagram of a forward biased and reverse biased p-n junction diode and explain their operation. 4
 (b) Draw the V-I characteristics curve of p-n diode and explain its nature. 3
 (c) The reverse saturation current of a p-n junction diode made of silicon is $8 \mu\text{A}/\text{m}^2$ at 300K. Find the voltage to be applied across the junction to obtain a forward current of 90 mA. 3

4. (a) Draw circuit diagram of bridge rectifier. Draw its input output waveforms and describe its operation. 5
 (b) Define PIV, ripple factor and efficiency of a rectifier. 3
 (c) In full wave bridge rectifier $V_m = 10\text{V}$, $R_L = 10 \text{ K}\Omega$ find out V_{DC} , I_{DC} 2

5. (a) Define α and β of transistor and also find the relation between them. 3
 (b) Draw O/P characteristics of CB configuration and explain its working. 4
 (c) Figure shows the voltage divider bias circuit. Determine the collector current and collector-to-emitter voltage. Assume $V_{BE} = 0.65$ and $\beta = 60$. Here $V_{CC} = 15\text{V}$, $R_1 = 10 \text{ K}\Omega$, $R_2 = 4.7 \text{ K}\Omega$, $R_c = 500 \Omega$ and $R_e = 350 \text{ K}\Omega$ 3



6. (a) Explain 'thermal runaway' of BJT. 2
 (b) What are the sources of instability of collector current in a transistor. Define three stability factors. 4
 (c) Draw the hybrid equivalent circuit of a generalized transistor amplifier and find current gain. 4

7. (a) Define feedback amplifier. Derive the overall gain of negative feedback amplifier. 4
 (b) What are the advantages of negative feedback amplifier? 3
 (c) Show that negative feedback improves the stability of gain of an amplifier. 3

8. (a) State the characteristics of an ideal OPAMP. 2
 (b) Define CMRR. Derive an expression for CMRR. Calculate the output voltage of a differential amplifier for the following parameters: $V_1 = 1000 \mu\text{V}$, $V_2 = 500 \mu\text{V}$, $A_d = 1000$ and $\text{CMRR} = 2000$ 5
 (c) Draw the circuit diagram of a non-inverting amplifier and obtain its gain. 3

9. (a) Explain the working principle of n-channel JFET. 4
 (b) Define the following parameters: i) transconductance ii) ac channel resistance iii) amplification factor. Deduce the relationship $\mu = r_d g_m$ 4
 (c) For an n-channel JFET $I_{DSS} = 8.7 \text{ mA}$, $V_P = -3\text{V}$, $V_{GS} = -1\text{V}$. Find the value of I_{DS} . 2

10. Write short note on any two.

- (a) Biased Clipping circuit (b) Zener diode as a voltage regulator
 (c) Transistor as a switch (d) Voltage Differentiator

5x: